

Week 2

```
mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
mysql> USE bookflow_db;
Reading table information for completion of table and column names
[You can turn off this feature to get a quicker startup with -A

Database changed
```

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
ERROR 1062 (23000): Duplicate entry '1' for key 'books.PRIMARY'
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018

```
3 rows in set (0.000 sec)
```

```
[mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (4, 'The Great Gatsby', '9780743273565', 1925),
  -> (5, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (6, '1984', '9780451524935', 1949);
Query OK, 3 rows affected (0.002 sec)
[Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132350884	2008
3	Atomic Habits	9780735211292	2018
4	The Great Gatsby	9780743273565	1925
5	To Kill a Mockingbird	9780061120084	1960
6	1984	9780451524935	1949

```
6 rows in set (0.000 sec)
```

```
mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
ERROR 1050 (42S01): Table 'members' already exists
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.001 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+
6 rows in set (0.000 sec)
```

```
mysql> CREATE TABLE Loans (
  -> loan_id INT PRIMARY KEY,
  -> member_id INT,
  -> book_id INT,
  -> loan_date DATE,
  -> FOREIGN KEY (member_id) REFERENCES Members(member_id),
  -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );
Query OK, 0 rows affected (0.013 sec)
```

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
  -> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
  -> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
  -> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
  -> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
  -> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
Query OK, 10 rows affected (0.004 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.000 sec)
```

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title  
-> FROM Loans l  
-> INNER JOIN Members m ON l.member_id = m.member_id  
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Alchemist
Emma Wilson	Clean Code
Michael Brown	Atomic Habits
John Smith	Clean Code
Emma Wilson	The Alchemist
Michael Brown	Clean Code
John Smith	Atomic Habits
Emma Wilson	Atomic Habits
Michael Brown	The Alchemist
John Smith	The Alchemist

```
10 rows in set (0.001 sec)
```

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1925	1
1949	1
1960	1
1988	1
2008	1
2018	1

```
6 rows in set (0.001 sec)
```

```
mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

```
Query OK, 0 rows affected (0.012 sec)
```

```
mysql> START TRANSACTION;
```

```
Query OK, 0 rows affected (0.000 sec)
```

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
```

```
ERROR 1062 (23000): Duplicate entry '4' for key 'books.PRIMARY'
```

```
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
```

```
Query OK, 1 row affected (0.002 sec)
```

```
mysql> COMMIT;
```

```
Query OK, 0 rows affected (0.001 sec)
```

```
mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
```

```
Query OK, 0 rows affected (0.017 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
```

book_id	title	isbn	published_year
6	1984	9780451524935	1949

```
1 row in set (0.001 sec)
```

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (7, 'Animal Farm', '9780451526342', 1945);
```

```
Query OK, 1 row affected (0.002 sec)
```

```
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
-> VALUES (4, 7, 'Raj Kumar', CURDATE());
```

```
Query OK, 1 row affected (0.002 sec)
```

```
mysql> COMMIT;CREATE INDEX idx_books_isbn ON Books(isbn);
```

```
Query OK, 0 rows affected (0.000 sec)
```

```
ERROR 1061 (42000): Duplicate key name 'idx_books_isbn'
```

```
mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
```

book_id	title	isbn	published_year
6	1984	9780451524935	1949

```
1 row in set (0.000 sec)
```